

Hein Zarni Aung

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EDUCATION

Stony Brook University

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Full Stack Development, Algorithms, Systems Fundamentals I & II

Expected Graduation: May 2027

INTERNSHIPS

Software Engineering Intern — Clustr

Cross-platform React Native social/scheduling app | ~8-person dev team

Jan 2026 - Sept 2026 (expected)

- Designed and implemented a Scheduler UI module featuring tabbed navigation (Suggestions, Requests, Responses), a reschedule bottom sheet, custom date/time picker integration, and a calendar-completion animation to improve user engagement.
- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows and activity feed endpoints.
- Implemented secure authentication features, including a password-reset flow (client screen plus backend cloud function) and custom claims to drive new-user onboarding logic.

Software Engineering Intern — Prenora Dynamics LLC

~8-person team

Jun 2026 - Sept 2026 (expected)

- Own end-to-end design decisions for a real-time, low-latency video chat feature built in-house, researching and evaluating SDK approaches to replace third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, and built a seed-data workflow to streamline debugging and local development.

CAMPUS INVOLVEMENT

Teaching Assistant — Data Structures & Algorithms, Stony Brook University

Aug 2025 - Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.

PROJECTS

TuneAcademy (React, NumPy, Essentia, FastAPI, Firestore)

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Pivoted from a planned machine learning model to rule-based scoring algorithms under hackathon time and compute constraints, delivering reliable weakness detection within the 36-hour build window.
- Set up a CI/CD pipeline using GitHub Actions to lint, test, and auto-deploy to Cloudflare Workers; broader app spans a React/Tailwind frontend, Python/FastAPI backend in Docker, Cloud Firestore, and GCP hosting.

SKILLS

Languages: Java, JavaScript, TypeScript, Python, C

Frontend: React, React Native, Next.js

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: MongoDB, AWS, Firebase/Firestore, PostgreSQL, GCP

Tools: Git, Docker, Linux, PM2, Selenium